NHANES Adult Audiometry 20–69 y: Combined-Cycle EDA

AUX1 (1999–2000) + AUX_G (2011–2012) + AUX_I (2015–2016)

2026-08-14

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Data Source and Files

Cycle	Exam file	Participants	Ages
AUX1 1999–2000	/opt/data/AUX1_9900.xpt	1,807	20–69

Cycle	Exam file	Participants	Ages
AUX_G 2011–2012	/opt/data/AUX_G_1112.xpt	4,500	20–69
AUX_I 2015–2016	/opt/data/AUX_I_1516.xpt	4,582	20–69

- **Protocol:** these cycles administered audiometry to adults aged 20–69 y (unlike the 2017–2020 P_AUX cycle, which tested only 6–19 and 70).
- **Frequencies:** 500, 1000, 2000, 3000, 4000, 6000, 8000 Hz, air-conduction, both ears.
- **Demographics:** linked via `SEQN` to each cycle's DEMO file (age, sex).

Cleaning

Sentinel codes converted to missing before any statistic: **666** (no response), **777** (refused), **888** (could not obtain), **999** (not tested). Values clipped to −10–120 dB HL. (The 1999–2000 cycle uses 666 only; 2011–2016 use 666 and 888.)

Sample Flow

Stage	Participants	Ears
Combined audiometry participants	10,889	—
Clean ears (all 7 freqs, no sentinels)	9,832	19,568

Ears per cycle: AUX_I 2015–2016 8,566, AUX_G 2011–2012 7,670, AUX1 1999–2000 3,332

Demographics

Age (full combined file): mean 43.9 y, median 44.0 y, range 20–69 y; 51.8% female.

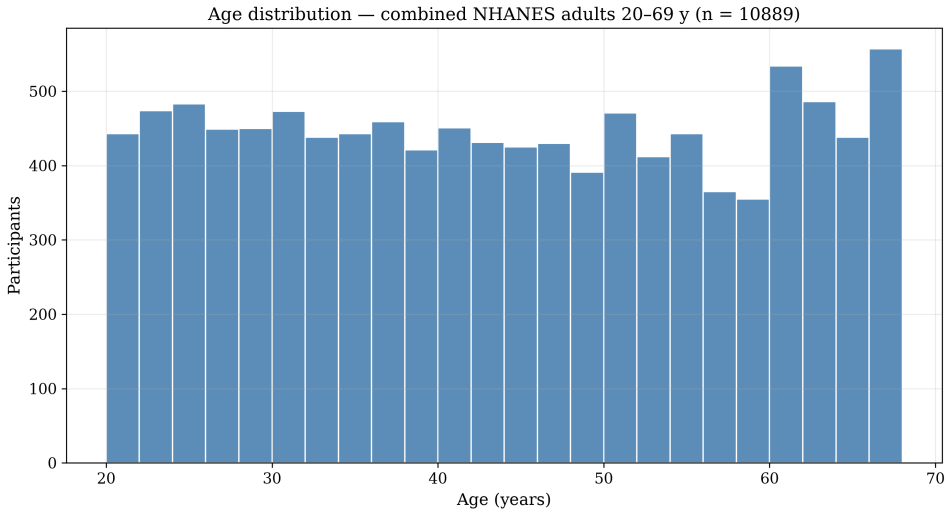

Figure 1: Age distribution

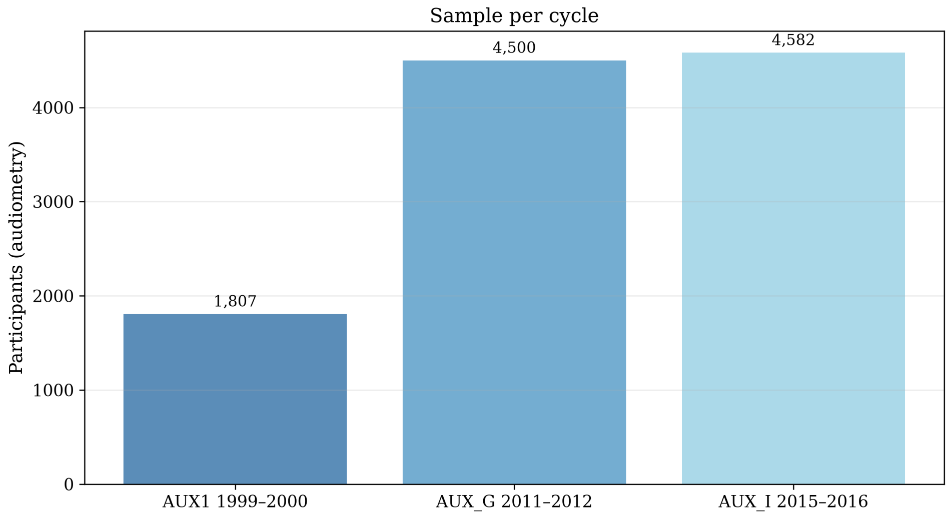

Figure 2: Sample per cycle

Clean-ear set: 9,832 participants, 19,568 ears, mean age 43.9 y, 50.9% female.

Threshold Distributions

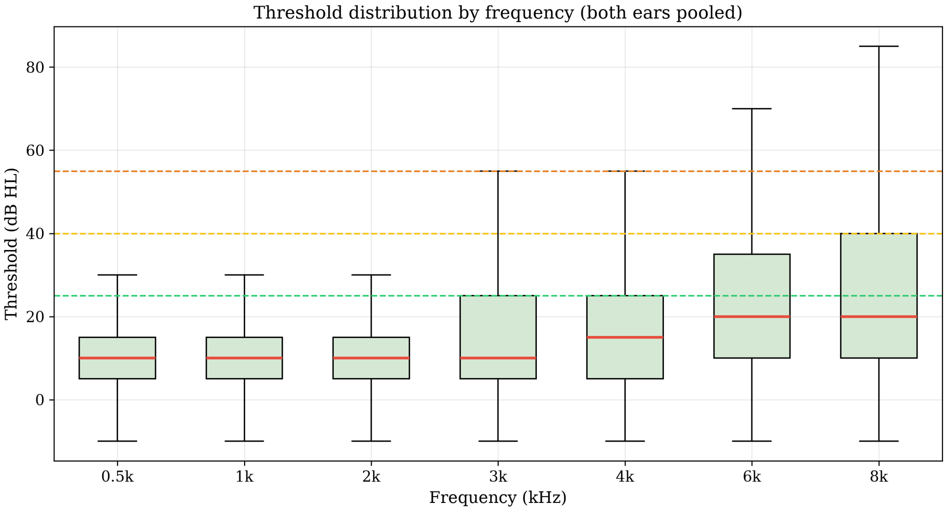

Figure 3: Threshold box plots by frequency

Frequency	n (pooled)	Mean (dB)	SD	Median
0.5k	19,655	11.2	10.8	10.0
1k	19,651	11.3	11.5	10.0
2k	19,651	11.9	13.8	10.0
3k	19,643	16.3	17.2	10.0
4k	19,634	19.5	19.3	15.0
6k	19,622	25.5	20.1	20.0
8k	19,589	27.4	22.4	20.0

PTA-4 and WHO Severity

PTA-4 = mean of 500, 1000, 2000, 4000 Hz.

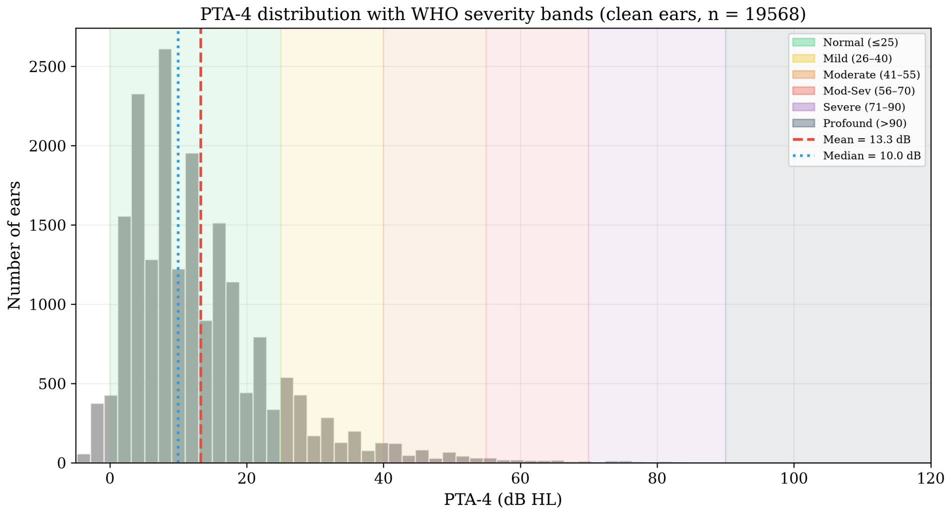

Figure 4: PTA-4 distribution with WHO severity bands

Severity (WHO)	% of clean ears
Normal	88.1%
Mild	8.7%
Moderate	2.2%
Moderately Severe	0.6%
Severe	0.3%
Profound	0.1%

The Borderline Problem

Share of clean ears within ±5 dB of a WHO severity boundary:

Boundary zone	dB range	% of ears
Normal–Mild	20–30 dB	13.9%
Mild–Moderate	35–45 dB	3.1%
Moderate–Mod-Sev	50–60 dB	0.9%
Mod-Sev–Severe	65–75 dB	0.3%
Severe–Profound	85–95 dB	0.0%

18.2% of clean ears fall within ±5 dB of a boundary.

Audiogram Configuration

Slope = threshold(4 kHz) − threshold(500 Hz).

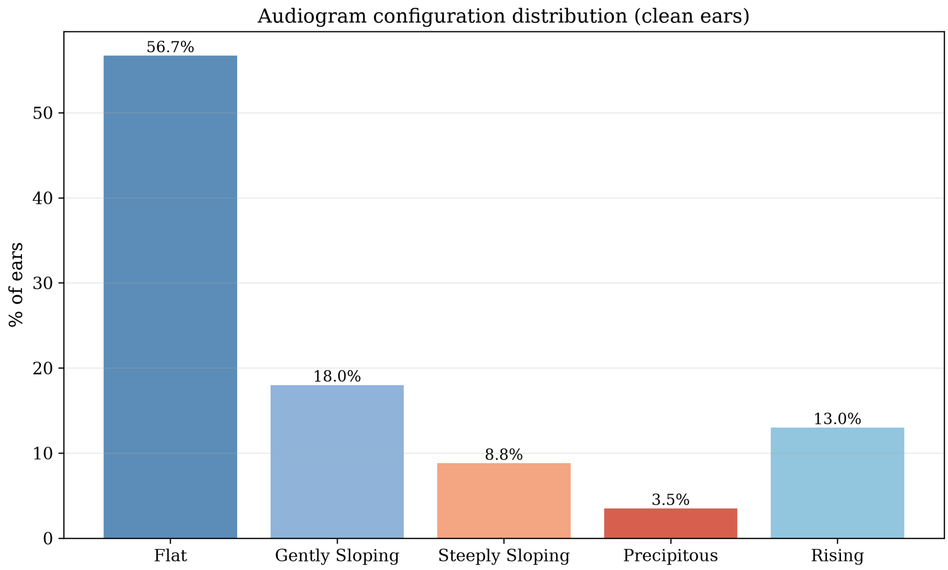

Figure 5: Configuration distribution

Configuration	% of clean ears
Flat	56.7%
Gently Sloping	18.0%
Steeply Sloping	8.8%
Precipitous	3.5%
Rising	13.0%

Inter-Aural Asymmetry

Bilateral participants (n = 9,736); |PTA-4 R − PTA-4 L|:

- 94.9% symmetric (15 dB)
- 3.5% mildly asymmetric (16–30 dB)
- 0.9% moderately asymmetric (31–45 dB)
- 0.6% severely asymmetric (>45 dB)

Mean asymmetry 5.2 dB, median 3.8 dB.

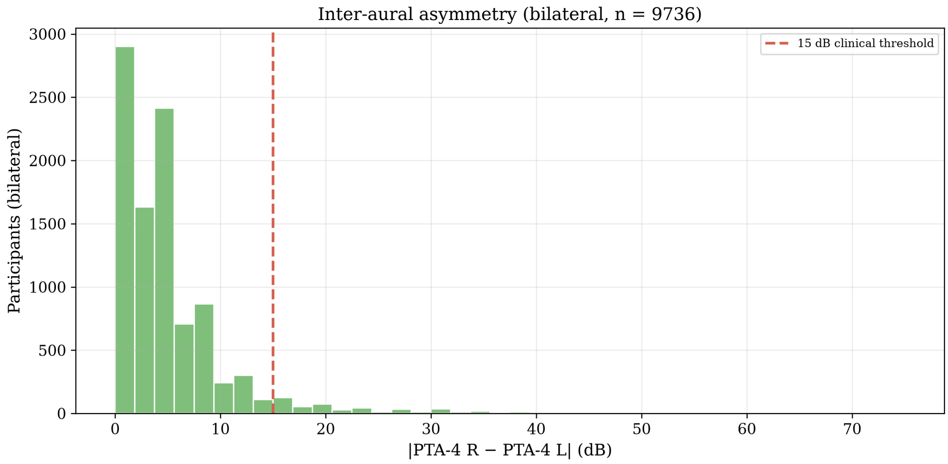

Figure 6: Inter-aural asymmetry

Inter-Frequency Correlations

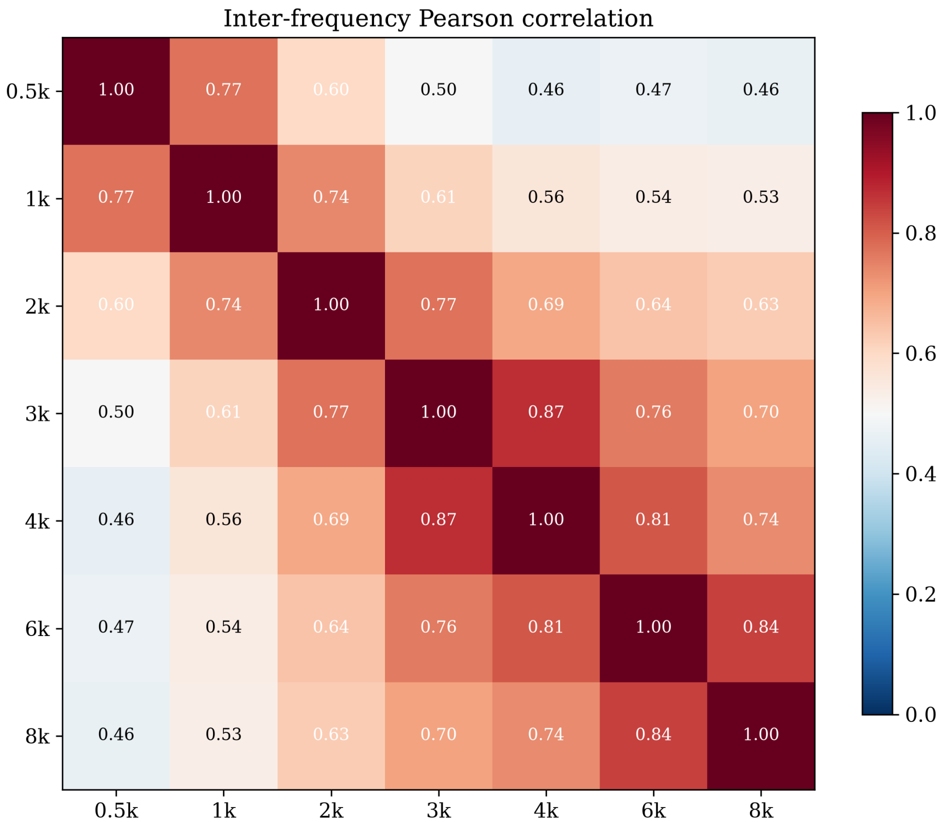

Figure 7: Correlation matrix

Adjacent-frequency mean r = 0.80; 500 Hz–8 kHz r = 0.46.

Missing Data Patterns

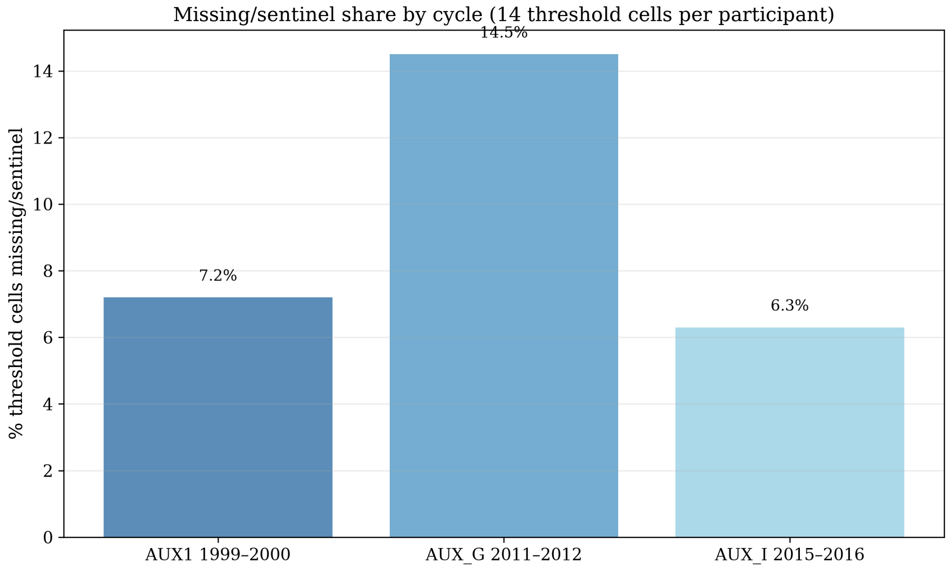

Figure 8: Missing/sentinel share by cycle

Key Takeaways

1. **True working-age adult cohort:** 10,889 participants aged 20–69 across three cycles — the population the 2017–2020 P_AUX cycle does not contain.
2. Hearing thresholds increase monotonically with frequency; absolute levels are lower than the 70 cohort (age-related loss less severe).
3. **18.2%** of ears sit within ±5 dB of a severity boundary — still a substantial borderline population.
4. Configurations are predominantly flat/gently sloping; steeply sloping and precipitous patterns are less common than in the 70 cohort.
5. Inter-frequency correlations justify frequency-resolved (fuzzy) representation.